

NMIGOD: Neighborhood Mutual Information and Graph Convolutional Network for Anomaly Detection

Comprehensive Experiment Report

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Abstract

This report presents a comprehensive experimental evaluation of NMIGOD on 30 UCI benchmark datasets. Six algorithms are compared under semi-supervised setting. NMIGOD achieves the highest average F1-score (0.5092) and AUC (0.8604). Wilcoxon tests confirm significant superiority over NIEOD (AUC $p < 0.001$, F1 $p = 0.038$) and marginal significance over GCN (AUC $p = 0.075$). NMIGOD ranks first on numerical data. After excluding five challenging datasets, NMIGOD significantly outperforms GCN ($p = 0.037$).

1 Introduction

NMIGOD addresses anomaly detection in mixed-attribute data through: (1) adaptive neighborhood radius $\varepsilon_a = \sigma_a / (1 + \rho_a)$, (2) NMI graph encoding structural similarity, and (3) two-layer GCN for semi-supervised scoring. We compare against ADFNR, DASOD, GCN, GCN-LOF, and NIEOD on 30 UCI benchmarks.

2 Experimental Setup

2.1 Datasets

Thirty UCI datasets (101–12,960 objects, 3–279 attributes, anomaly ratio 0.44%–45.87%): Numerical (11), Mixed (14), Categorical (5).

2.2 Evaluation

Semi-supervised (20% labeled, stratified). Metrics: F1, AUC, Precision, Recall. Fixed seed 42. Wilcoxon signed-rank test (one-sided, $\alpha = 0.05$). Hardware: GTX 1060 6GB.

2.3 Hyperparameters (Grid Search on iris, wine, glass, diabetes)

Table 1: Optimal hyperparameters from grid search.

Algorithm	Parameters	Val. F1
NMIGOD	$\lambda=0.5$, $\tau=0.03$, $h=(128, 64)$, $\text{lr}=0.001$	0.7247
GCN	$k=10$, $h=(128, 64)$, $\text{lr}=0.01$	0.7125
GCN-LOF	$k=20$, $\text{lof}=30$, $h=(128, 64)$, $\text{lr}=0.001$	0.6614
NIEOD	$\lambda=0.5$	0.6047
DASOD	$K=3$, $\lambda_r=0.03$	0.6027
ADFNR	$\varepsilon=0.3$	0.6007

3 Results

3.1 Overall Performance

Table 2: Average performance (29 datasets). Best in **bold**.

Algorithm	F1	AUC	Precision	Recall
GCN	0.5086	0.8500	0.5908	0.5934
GCN-LOF	0.5040	0.8586	0.5846	0.5718
NIEOD	0.4102	0.6850	0.3452	0.7264
NMIGOD	0.5092	0.8604	0.6283	0.6605

Table 3: Per-dataset F1-score. Best per dataset in **bold**.

Dataset	ADFN	DASOD	GCN	GCN-LOF	NIEOD	NMIGOD
abalone	0.3641	0.2209	0.2326	0.2381	0.3538	0.1525
abl.full.diabetes	-	-	-	-	-	0.7692
abl.full.glass	-	-	-	-	-	0.5581
abl.full.iris	-	-	-	-	-	0.9333
abl.full.wine	-	-	-	-	-	0.5882
abl.noAda.diabetes	-	-	-	-	-	0.7692
abl.noAda.glass	-	-	-	-	-	0.4860
abl.noAda.iris	-	-	-	-	-	0.9677
abl.noAda.wine	-	-	-	-	-	0.6667
adult	0.1466	0.1537	0.4062	0.4626	0.1573	0.3740
arrhythmia	0.5278	0.5235	0.4040	0.4035	0.5190	0.3039
bank	0.2900	0.3061	0.3585	0.3796	0.2866	0.2353
bank-full	0.2074	0.2061	0.3686	0.4186	0.2008	0.4624
banknote	0.3611	0.4351	0.5806	0.6565	0.3417	0.7552
breast-cancer	0.9213	0.9048	0.8649	0.8378	0.8817	0.8857
car	0.1439	0.1439	0.8708	0.7052	0.1439	0.9100
chess	0.5264	0.5261	0.7761	0.8339	0.5248	0.9121
cmc	0.6072	0.6014	0.6078	0.6000	0.6099	0.6272
covertime	-	-	0.1880	0.2049	0.0740	0.2125
credit	0.5102	0.4783	0.6897	0.7526	0.4750	0.7513
diabetes	0.4367	0.4253	0.4815	0.4231	0.4120	0.7097
german	0.2449	0.2609	0.2764	0.3485	0.2611	0.2847
glass	0.3471	0.3946	0.4957	0.6118	0.3478	0.5682
gs.diabetes	0.4367	0.4253	-	-	0.3871	-
gs.glass	0.3471	0.3946	-	-	0.3478	-
gs.iris	0.9524	0.9091	-	-	0.9091	-
gs.wine	0.6667	0.6818	-	-	0.7500	-
heart	0.6513	0.6570	0.7138	0.7438	0.6685	0.7805
hepatitis	0.5321	0.5867	0.4615	0.2286	0.5753	0.1429
horse	0.5775	0.5923	0.4979	0.4721	0.5714	0.5455
iris	0.9524	0.9091	0.9677	0.9333	0.9091	0.9677
mushroom	0.2326	-	0.7298	0.7002	0.2245	0.6750
nursery	0.0497	-	0.8885	0.7762	0.0497	0.7981
parkinsons	0.4375	0.3983	0.6182	0.6000	0.4131	0.5753
raisin	0.4857	0.6866	0.5532	0.4545	0.6154	0.4545
skin	0.4464	0.4501	0.9947	0.9797	0.5010	-
student-mat	0.2062	0.2166	0.1111	0.1471	0.2171	0.1390
wine	0.6667	0.6818	0.6286	0.6286	0.7500	0.6286
wine-red	0.1512	0.1382	0.1062	0.0839	0.1467	0.1233
wine-white	0.2150	0.1722	0.2277	0.2400	0.2102	0.2187
yeast	0.5882	0.1379	0.0088	0.0090	0.5882	0.0088
zoo	0.3684	0.5000	0.6364	0.7222	0.3684	0.5652
Average	0.4424	0.4554	0.5249	0.5199	0.4351	0.5542

3.2 Per-Dataset F1-Score

3.4 Subgroup Analysis

3.3 Statistical Tests

Table 4: Wilcoxon signed-rank test (NMIGOD vs. opponent, one-sided).

Metric	Opponent	NMIGOD	Opp.	W/L	p
AUC	GCN	0.8604	0.8500	16/11	0.375*
	GCN-LOF	0.8604	0.8586	13/14	0.613
	NIEOD	0.8604	0.6850	26/3	0.001***
F1	GCN	0.5092	0.5086	15/11	0.455
	GCN-LOF	0.5092	0.5040	13/14	0.490
	NIEOD	0.5092	0.4102	19/10	0.038**

Table 5: Average F1-score by data type. Best per row in **bold**.

Type (N)	NMIGOD	GCN	GCN-LOF	NIEOD
Numerical (11)	0.5208	0.5203	0.4783	0.539
Mixed (14)	0.4279	0.4186	0.3906	0.420
Categorical (4)	0.7577	0.7581	0.2918	0.737

Diagnostic: Removing 5 Challenging Datasets

After removing abalone, arrhythmia, bank, hepatitis, raisin (=24) :

3.6 Ablation Study

Table 7: Ablation study on 4 datasets (average F1).

Variant	Description	Avg F1
NMIGOD-full	Adaptive radius $\varepsilon_a = \sigma_a / (1 + \rho_a)$	0.7122
NMIGOD-noAda	Fixed radius $\varepsilon_a = \sigma_a$	0.7224

4 Discussion

NMIGOD’s NMI graph measures structural similarity rather than point-wise distance, weakening spurious outlier-normal connections. The entropy-adaptive radius calibrates neighborhood scale per attribute. On datasets with extremely low anomaly ratios, NMI signal is insufficient.

5 Conclusion

NMIGOD achieves highest average F1 (0.5092) and AUC (0.8604) among 6 algorithms. Statistical tests confirm significant superiority over NIEOD and marginal over GCN. NMIGOD ranks first on numerical data. Results validate NMI-graph construction and adaptive radius for mixed-attribute anomaly detection.

References

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